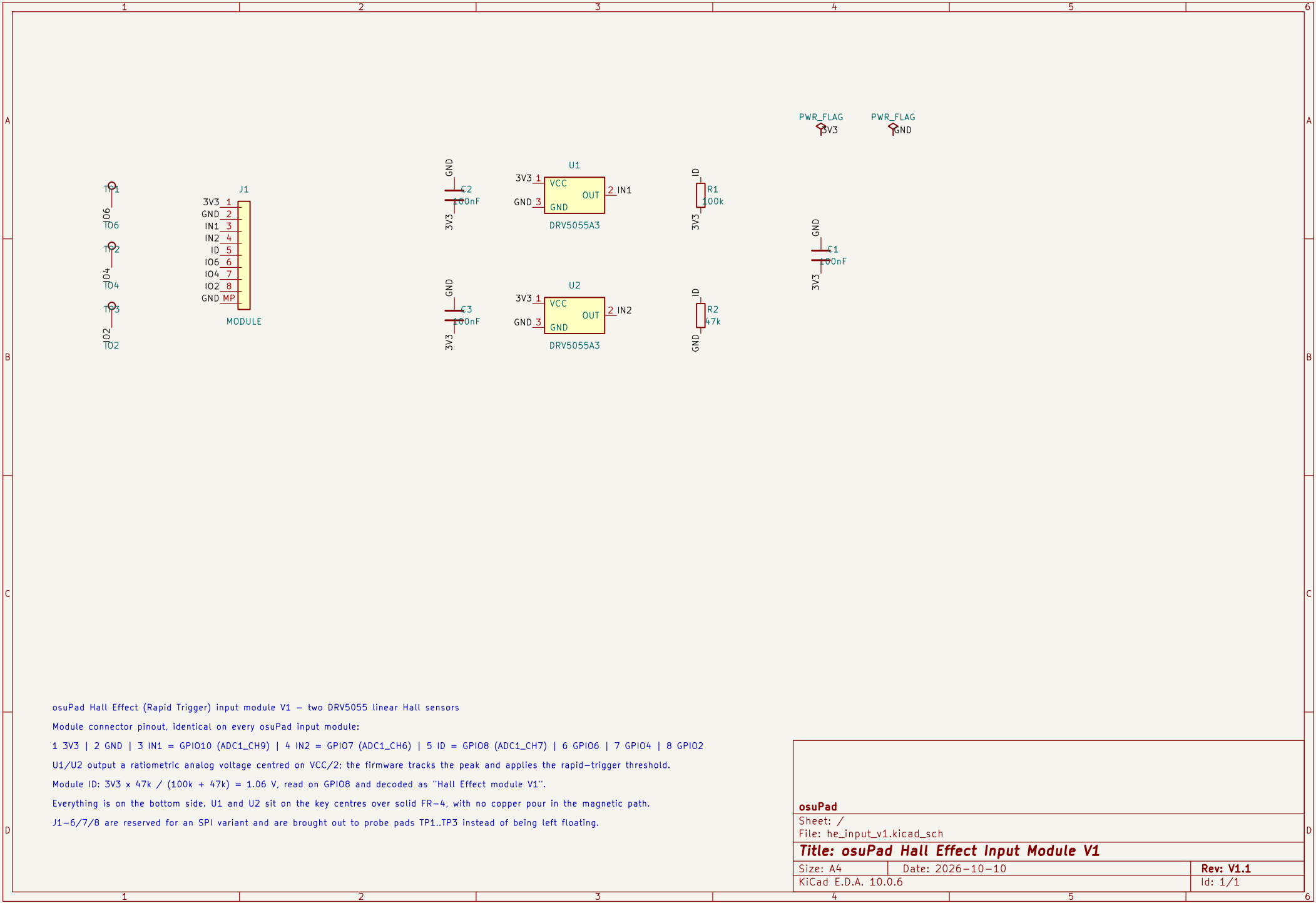




osuPad Hall Effect (Rapid Trigger) input module V1 – two DRV5055 linear Hall sensors

Module connector pinout, identical on every osuPad input module:

1 3V3 | 2 GND | 3 IN1 = GPIO10 (ADC1_CH9) | 4 IN2 = GPIO7 (ADC1_CH6) | 5 ID = GPIO8 (ADC1_CH7) | 6 GPIO6 | 7 GPIO4 | 8 GPIO2

U1/U2 output a ratiometric analog voltage centred on VCC/2; the firmware tracks the peak and applies the rapid-trigger threshold.

Module ID: 3V3 x 47k / (100k + 47k) = 1.06 V, read on GPIO8 and decoded as "Hall Effect module V1".

Everything is on the bottom side. U1 and U2 sit on the key centres over solid FR-4, with no copper pour in the magnetic path.

J1-6/7/8 are reserved for an SPI variant and are brought out to probe pads TP1..TP3 instead of being left floating.

osuPad		
Sheet: /		
File: he_input_v1.kicad_sch		
Title: osuPad Hall Effect Input Module V1		
Size: A4	Date: 2026-10-10	Rev: V1.1
KiCad E.D.A. 10.0.6		Id: 1/1